

Cr. No. 784/2026, PS Shamshabad Rural (synthetic)

|                  |                                                             |                  |                            |
|------------------|-------------------------------------------------------------|------------------|----------------------------|
| Title            | Roadside homicide with burning — culvert under service road | LLM layer        | none                       |
| Crime type       | Homicide / suspicious death                                 | Case ID          | PRM-20260928-B4ECB         |
| Exhibits         | 18 total - P1 8 / P2 8 / P3 1 / P4 1                        | Generated (UTC)  | 28 Sep 2026 06:47          |
| First submission | 17 exhibits (stage 1)                                       | Custody deadline | 25 Dec 2026 (BNSS §187(3)) |

1. Forwarding note

The exhibits listed in the enclosed schedule, seized in connection with Cr. No. 784/2026, PS Shamshabad Rural (synthetic) (homicide / suspicious death), are forwarded for scientific examination. Of 18 exhibits, 8 are marked Critical (P1) and 8 High (P2); 8 carry urgent-testing flags. Exhibits requiring immediate attention: Ex-A5 (accelerant container (petrol / kerosene): perishable); Ex-A17 (post-mortem viscera / stomach contents: perishable); Ex-A12 (footwear impression: collect now); Ex-A13 (cctv footage / dvr / nvr: digital overwrite). An accused is in judicial/police custody; reports are requested before 25 Dec 2026 to enable filing of the final report within the statutory period. Priorities were computed by the Pramaan engine v1.0.0 (knowledge base 858d6dd5797d) and are submitted for the Investigating Officer's review; the FSL may re-order examinations on scientific grounds.

2. ACT NOW - field and preservation actions

| Exhibit | Type                                     | Action                                                                                                                                                                          |
|---------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ex-A5   | Accelerant container (petrol / kerosene) | PERISHABLE: Degrades below usable quality in ~42 h under ambient storage. Move to an airtight container (unused paint can / nylon fire-debris bag) to extend this to ~104 days. |
| Ex-A17  | Post-mortem viscera / stomach contents   | PERISHABLE: Degrades below usable quality in ~42 h under ambient storage. Move to refrigeration (2-8 °C) to extend this to ~17 days.                                            |
| Ex-A12  | Footwear impression                      | COLLECT_NOW: Still at the scene: evidence is likely lost in ~36 h (overwrite, weathering, washing). Collect immediately.                                                        |
| Ex-A13  | CCTV footage / DVR / NVR                 | DIGITAL_OVERWRITE: DVR/NVR storage loops and overwrites (often 7-30 days). Seize or export with hash today.                                                                     |
| Ex-A11  | Tyre impression / tread mark             | COLLECT_NOW: Still at the scene: evidence is likely lost in ~36 h (overwrite, weathering, washing). Collect immediately.                                                        |
| Ex-A6   | Fire debris / burnt material             | PERISHABLE: Degrades below usable quality in ~42 h under ambient storage. Move to an airtight container (unused paint can / nylon fire-debris bag) to extend this to ~104 days. |

3. Ranked examination schedule (stage 1)

| # | Exhibit | Description                                           | Type                                     | Tier | EPI  | Examination sequence                                                                                                                                                                     | Urgent flags                   |
|---|---------|-------------------------------------------------------|------------------------------------------|------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| 1 | Ex-A8   | bloodstained stone approx 2 kg near the body          | Blunt object used as weapon              | P1   | 61.8 | Trace recovery (visual search, taping for hair/fibre) -> Presumptive blood test & species origin -> Touch-DNA swabbing of handling zones -> DNA extraction, quantitation & STR profiling | -                              |
| 2 | Ex-A2   | half-burnt cigarette butt, 3 m from the body          | Cigarette / beedi butt                   | P1   | 61.6 | DNA extraction, quantitation & STR profiling -> Fire debris / accelerant identification (GC-MS)                                                                                          | -                              |
| 3 | Ex-A5   | empty petrol bottle (1 litre) found 5 m from the body | Accelerant container (petrol / kerosene) | P1   | 57.3 | Latent print development (powder / cyanoacrylate / ninhydrin) -> Fire debris / accelerant identification (GC-MS) -> Latent print comparison & AFIS search                                | PERISHABLE, VOLATILE_TOXICANTS |

| #  | Exhibit       | Description                                                                               | Type                                                                 | Tier      | EPI  | Examination sequence                                                                                                                                                                                                                                                                                     | Urgent flags                                        |
|----|---------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-----------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| 4  | <b>Ex-A17</b> | viscera preserved at autopsy, kept at ambient temperature                                 | Post-mortem viscera / stomach contents                               | <b>P1</b> | 54.3 | Volatile analysis (ethanol, cyanide, phosphine — headspace GC) -> Toxicology screening (drugs, pesticides, poisons)                                                                                                                                                                                      | PERISHABLE, VOLATILE_TOXICANTS, COLD_CHAIN_REQUIRED |
| 5  | <b>Ex-A12</b> | footprints near the culvert, not yet collected                                            | Footwear impression                                                  | <b>P1</b> | 51.8 | Footwear impression comparison                                                                                                                                                                                                                                                                           | COLLECT_NOW                                         |
| 6  | <b>Ex-A13</b> | CCTV at toll plaza covering the service road, not yet seized                              | CCTV footage / DVR / NVR                                             | <b>P1</b> | 51.5 | CCTV / DVR footage extraction & hash verification                                                                                                                                                                                                                                                        | DIGITAL_OVERWRITE, BSA63_CERTIFICATE_PENDING        |
| 7  | <b>Ex-A11</b> | tyre impressions of heavy vehicle on soft soil, not yet collected                         | Tyre impression / tread mark                                         | <b>P1</b> | 49.5 | Tyre impression comparison                                                                                                                                                                                                                                                                               | COLLECT_NOW                                         |
| 8  | <b>Ex-A6</b>  | burnt cloth pieces and ash sample from the burn area                                      | Fire debris / burnt material                                         | <b>P1</b> | 48.6 | Fire debris / accelerant identification (GC-MS)                                                                                                                                                                                                                                                          | PERISHABLE, VOLATILE_TOXICANTS                      |
| 9  | <b>Ex-A3</b>  | 6 cigarette butts of the same brand near the parked lorry, 40 m from the body             | Cigarette / beedi butt                                               | <b>P2</b> | 59.6 | DNA extraction, quantitation & STR profiling                                                                                                                                                                                                                                                             | -                                                   |
| 10 | <b>Ex-A4</b>  | two empty liquor bottles beside the lorry parking area                                    | Bottle / glass / cup (saliva + prints)                               | <b>P2</b> | 56.7 | Touch-DNA swabbing of handling zones -> Latent print development (powder / cyanoacrylate / ninhydrin) -> DNA extraction, quantitation & STR profiling -> Latent print comparison & AFIS search                                                                                                           | -                                                   |
| 11 | <b>Ex-A18</b> | lighter recovered from the accused                                                        | Litter handled by offender (gutka / food wrapper, matchbox, lighter) | <b>P2</b> | 56.4 | Touch-DNA swabbing of handling zones -> Latent print development (powder / cyanoacrylate / ninhydrin) -> DNA extraction, quantitation & STR profiling -> Latent print comparison & AFIS search                                                                                                           | -                                                   |
| 12 | <b>Ex-A15</b> | clothes of accused seized at the instance of accused                                      | Clothing / bedding / textiles                                        | <b>P2</b> | 55.7 | Trace recovery (visual search, taping for hair/fibre) -> Presumptive blood test & species origin -> Body-fluid identification (semen / saliva) -> DNA extraction, quantitation & STR profiling -> Fibre comparison (microscopy / FTIR)                                                                   | -                                                   |
| 13 | <b>Ex-A1</b>  | charred remains recovered under the culvert                                               | Charred remains / bone / teeth                                       | <b>P2</b> | 54.1 | DNA from bone / tooth (demineralisation) -> Fire debris / accelerant identification (GC-MS)                                                                                                                                                                                                              | -                                                   |
| 14 | <b>Ex-A14</b> | 4 mobile phones seized from the accused                                                   | Mobile phone / SIM                                                   | <b>P2</b> | 52.1 | Mobile device extraction (logical / physical)                                                                                                                                                                                                                                                            | DIGITAL_VOLATILE                                    |
| 15 | <b>Ex-A10</b> | gutka packet and matchbox near the culvert entrance                                       | Litter handled by offender (gutka / food wrapper, matchbox, lighter) | <b>P2</b> | 50.3 | Touch-DNA swabbing of handling zones -> Latent print development (powder / cyanoacrylate / ninhydrin) -> DNA extraction, quantitation & STR profiling -> Latent print comparison & AFIS search                                                                                                           | -                                                   |
| 16 | <b>Ex-A9</b>  | mobile phone of the deceased, switched off, found in bushes                               | Mobile phone / SIM                                                   | <b>P2</b> | 47.9 | Mobile device extraction (logical / physical)                                                                                                                                                                                                                                                            | DIGITAL_VOLATILE                                    |
| 17 | <b>Ex-A7</b>  | two-wheeler (scooty) of the deceased with deflated rear tyre, parked near toll plaza road | Vehicle (two-wheeler, car, truck)                                    | <b>P3</b> | 40.5 | Trace recovery (visual search, taping for hair/fibre) -> Presumptive blood test & species origin -> Latent print development (powder / cyanoacrylate / ninhydrin) -> DNA extraction, quantitation & STR profiling -> Glass / paint physical-chemical comparison -> Latent print comparison & AFIS search | -                                                   |

Tiers: P1 Critical — examine immediately; P2 High — examine in first batch; P3 Routine — standard queue; P4 Hold — store; examine only if case theory needs it.

## 4. Handling, caveats and corroboration (P1-P2)

### Ex-A8 - Blunt object used as weapon (P1, EPI 61.8)

Why: Blunt object used as weapon: relevance 1.00 for homicide / suspicious death (weapon evidence), individualizing potential 0.80; modifiers: blood; context: near\_body (+0.15); usable quality remaining ~22 days under ambient storage; irreplaceability 1.00. EPI 61.8.

Handling: Rigid box; protect stained surfaces; air-dry if wet.

Does not prove: A bloodstained object near a body is consistent with use as a weapon; confirm with injury pattern.

Corroborate with: Post-mortem injury pattern; Hair/tissue adhering to object

### Ex-A2 - Cigarette / beedi butt (P1, EPI 61.6)

Why: Cigarette / beedi butt: relevance 1.00 for homicide / suspicious death (biological evidence), individualizing potential 0.80; modifiers: burnt; context: near\_body, 3 m from body/point of offence (+0.13); usable quality remaining ~14 days under ambient storage; irreplaceability 1.00. EPI 61.6.

Handling: Pick with forceps by the burnt end, air-dry, pack EACH butt separately in paper. Photograph in situ with scale first.

Does not prove: Proves a smoker's saliva DNA was deposited — not when. Judge freshness (moisture, ash, brand) and background litter before inferring presence at the time of offence.

Corroborate with: Brand found with suspects; CCTV at nearby shops (purchase); Other butts of the same brand nearby

HEAT\_DAMAGED\_DNA (medium): Heat-exposed biological material: sample protected areas (seams, folds); expect partial profiles.

### Ex-A5 - Accelerant container (petrol / kerosene) (P1, EPI 57.3)

Why: Accelerant container (petrol / kerosene): relevance 0.80 for homicide / suspicious death (chemical evidence), individualizing potential 0.65; context: near\_body, 5 m from body/point of offence (+0.12); usable quality remaining ~42 h under ambient storage; irreplaceability 1.00. EPI 57.3.

Handling: Airtight metal can or nylon fire-debris bag — never polythene; keep cool.

Does not prove: Accelerant identity (e.g., petrol) supports deliberate ignition; linking the container to a person needs prints/DNA or purchase evidence.

Corroborate with: Fire debris analysis; Fuel station CCTV / receipts; Latent prints

PERISHABLE (critical): Degrades below usable quality in ~42 h under ambient storage. Move to an airtight container (unused paint can / nylon fire-debris bag) to extend this to ~104 days.

VOLATILE\_TOXICANTS (high): Volatile compounds (alcohol, cyanide, phosphine, accelerants) are lost with delay — request headspace/volatile analysis first.

### Ex-A17 - Post-mortem viscera / stomach contents (P1, EPI 54.3)

Why: Post-mortem viscera / stomach contents: relevance 0.80 for homicide / suspicious death (toxicology evidence), individualizing potential 0.70; usable quality remaining ~42 h under ambient storage; irreplaceability 1.00. EPI 54.3.

Handling: Preserve in saturated salt solution (not for alcohol/volatile cases) in sealed glass jars; refrigerate; forward urgently.

Does not prove: A negative screen does not exclude poisoning (unscreened compounds, decomposition, delay).

Corroborate with: Post-mortem findings; Suspected poison container; Treating hospital records

PERISHABLE (critical): Degrades below usable quality in ~42 h under ambient storage. Move to refrigeration (2–8 °C) to extend this to ~17 days.

VOLATILE\_TOXICANTS (high): Volatile compounds (alcohol, cyanide, phosphine, accelerants) are lost with delay — request headspace/volatile analysis first.

COLD\_CHAIN\_REQUIRED (high): Maintain cold chain (2–8 °C) from collection to FSL receipt; record temperatures.

### Ex-A12 - Footwear impression (P1, EPI 51.8)

Why: Footwear impression: relevance 0.80 for homicide / suspicious death (impression evidence), individualizing potential 0.55; usable quality remaining ~36 h under ambient storage; irreplaceability 1.00. EPI 51.8.

Handling: Photograph with scale at 90°, then cast (dental stone). Protect from rain immediately.

Does not prove: Sole pattern narrows the footwear model; individualization needs accidental characteristics (cuts, wear).

Corroborate with: Suspect footwear; Direction of travel; Number of distinct patterns (number of persons)

COLLECT\_NOW (critical): Still at the scene: evidence is likely lost in ~36 h (overwrite, weathering, washing). Collect immediately.

### **Ex-A13 - CCTV footage / DVR / NVR (P1, EPI 51.5)**

Why: CCTV footage / DVR / NVR: relevance 0.85 for homicide / suspicious death (digital evidence), individualizing potential 0.80; usable quality remaining ~7 days under ambient storage; irreplaceability 1.00. EPI 51.5.

Handling: Seize DVR or export native format with hash value on the spot; note clock offset; certificate under BSA 2023 §63.

Does not prove: Footage timestamps may be offset from real time — record the DVR clock offset at seizure.

Corroborate with: Other cameras on the route; Vehicle registration databases; Mobile tower data

DIGITAL\_OVERWRITE (critical): DVR/NVR storage loops and overwrites (often 7–30 days). Seize or export with hash today.

BSA63\_CERTIFICATE\_PENDING (high): s.63(4) certificate (Schedule Part A + Part B with hash) must accompany this record every time it is submitted.

### **Ex-A11 - Tyre impression / tread mark (P1, EPI 49.5)**

Why: Tyre impression / tread mark: relevance 0.80 for homicide / suspicious death (impression evidence), individualizing potential 0.45; usable quality remaining ~36 h under ambient storage; irreplaceability 1.00. EPI 49.5.

Handling: Photograph full circumference length with scale; cast; record track width.

Does not prove: Tread design is class evidence; measure track width and wheelbase to narrow vehicle type.

Corroborate with: Suspect vehicle tyres; CCTV of vehicles passing; Paint / glass transfer

COLLECT\_NOW (critical): Still at the scene: evidence is likely lost in ~36 h (overwrite, weathering, washing). Collect immediately.

### **Ex-A6 - Fire debris / burnt material (P1, EPI 48.6)**

Why: Fire debris / burnt material: relevance 0.80 for homicide / suspicious death (chemical evidence), individualizing potential 0.45; modifiers: burnt; usable quality remaining ~42 h under ambient storage; irreplaceability 1.00. EPI 48.6.

Handling: Airtight container filled to 2/3, collected from low points of burn pattern; include an unburnt control.

Does not prove: Accelerant residues evaporate within hours to days; a negative result after delay is weak evidence of absence.

Corroborate with: Burn pattern photographs; Accelerant container; Comparison sample from unburnt area

PERISHABLE (critical): Degrades below usable quality in ~42 h under ambient storage. Move to an airtight container (unused paint can / nylon fire-debris bag) to extend this to ~104 days.

VOLATILE\_TOXICANTS (high): Volatile compounds (alcohol, cyanide, phosphine, accelerants) are lost with delay — request headspace/volatile analysis first.

### **Ex-A3 - Cigarette / beedi butt (P2, EPI 59.6)**

Why: Cigarette / beedi butt: relevance 1.00 for homicide / suspicious death (biological evidence), individualizing potential 0.90; context: 40 m from body/point of offence (+0.03); usable quality remaining ~14 days under ambient storage; irreplaceability 1.00. EPI 59.6.

Handling: Pick with forceps by the burnt end, air-dry, pack EACH butt separately in paper. Photograph in situ with scale first.

Does not prove: Proves a smoker's saliva DNA was deposited — not when. Judge freshness (moisture, ash, brand) and background litter before inferring presence at the time of offence.

Corroborate with: Brand found with suspects; CCTV at nearby shops (purchase); Other butts of the same brand nearby

### **Ex-A4 - Bottle / glass / cup (saliva + prints) (P2, EPI 56.7)**

Why: Bottle / glass / cup (saliva + prints): relevance 1.00 for homicide / suspicious death (biological evidence), individualizing potential 0.85; usable quality remaining ~14 days under ambient storage; irreplaceability 1.00. EPI 56.7.

Handling: Hold at base with gloved fingers; swab rim before any fingerprint powder; pack upright in a box, not a bag.

Does not prove: Rim DNA and prints show handling/drinking by a person, not when. Shared bottles often give mixed profiles.

Corroborate with: Remaining liquid (toxicology if drugging suspected); Shop CCTV / bill

SEQUENCE\_DNA\_BEFORE\_PRINTS (medium): Needs both DNA and latent-print work: swab handling zones for touch DNA before powder/cyanoacrylate enhancement; route DNA ↔ Fingerprint in the sequence shown.

#### **Ex-A18 - Litter handled by offender (gutka / food wrapper, matchbox, lighter) (P2, EPI 56.4)**

Why: Litter handled by offender (gutka / food wrapper, matchbox, lighter): relevance 1.00 for homicide / suspicious death (biological evidence), individualizing potential 0.60; context: suspect\_link (+0.15); usable quality remaining ~10 days under ambient storage; irreplaceability 1.00. EPI 56.4.

Handling: Paper envelope, do not fold across print areas, avoid plastic if damp.

Does not prove: Touch DNA on handled litter is often low-template and may be mixed; background litter is common at roadsides.

Corroborate with: Brand/batch found with suspects; Proximity to body or point of entry

SEQUENCE\_DNA\_BEFORE\_PRINTS (medium): Needs both DNA and latent-print work: swab handling zones for touch DNA before powder/cyanoacrylate enhancement; route DNA ↔ Fingerprint in the sequence shown.

#### **Ex-A15 - Clothing / bedding / textiles (P2, EPI 55.7)**

Why: Clothing / bedding / textiles: relevance 0.90 for homicide / suspicious death (clothing evidence), individualizing potential 0.75; context: suspect\_link (+0.15); usable quality remaining ~22 days under ambient storage; irreplaceability 1.00. EPI 55.7.

Handling: Air-dry in shade on clean paper, fold with paper between layers, EACH garment in a separate paper bag.

Does not prove: Stains on clothing show transfer, not the mechanism. Wearer's own DNA will be present; interpret foreign profiles only.

Corroborate with: Bloodstain pattern on garment; Damage (cuts/tears) vs weapon; Wearer reference sample

#### **Ex-A1 - Charred remains / bone / teeth (P2, EPI 54.1)**

Why: Charred remains / bone / teeth: relevance 1.00 for homicide / suspicious death (biological evidence), individualizing potential 0.85; modifiers: burnt; usable quality remaining ~269 days under ambient storage; irreplaceability 0.95. EPI 54.1.

Handling: Collect with forensic odontologist/anthropologist where possible; dry, rigid container, no plastic.

Does not prove: Establishes identity of the deceased; heat damage may yield partial profiles. Long turnaround — start early.

Corroborate with: Reference samples from biological parents/children; Dental records; Personal effects

HEAT\_DAMAGED\_DNA (medium): Heat-exposed biological material: sample protected areas (seams, folds); expect partial profiles.

#### **Ex-A14 - Mobile phone / SIM (P2, EPI 52.1)**

Why: Mobile phone / SIM: relevance 0.85 for homicide / suspicious death (digital evidence), individualizing potential 0.85; context: suspect\_link (+0.15); physically stable in storage; irreplaceability 0.80. EPI 52.1.

Handling: Airplane mode if unlocked, otherwise Faraday bag; keep charged; do NOT browse the device; record IMEI.

Does not prove: Device data shows what the device did; attributing actions to a person needs corroboration. Certificate under BSA 2023 §63 needed for admissibility.

Corroborate with: CDR / tower location from service provider; CCTV; Cloud account records

DIGITAL\_VOLATILE (high): Isolate from networks (Faraday bag / airplane mode), keep powered; remote wipe and lock-out risk.

BSA63\_DETAILS\_MISSING (medium): Electronic record: record acquisition method, hash (SHA-256), s.63(2) facts and certificate status — needed to prove it under BSA 2023 s.63.

#### **Ex-A10 - Litter handled by offender (gutka / food wrapper, matchbox, lighter) (P2, EPI 50.3)**

Why: Litter handled by offender (gutka / food wrapper, matchbox, lighter): relevance 1.00 for homicide / suspicious death (biological evidence), individualizing potential 0.60; usable quality remaining ~10 days under ambient storage; irreplaceability 1.00. EPI 50.3.

Handling: Paper envelope, do not fold across print areas, avoid plastic if damp.

Does not prove: Touch DNA on handled litter is often low-template and may be mixed; background litter is common at roadsides.

Corroborate with: Brand/batch found with suspects; Proximity to body or point of entry

SEQUENCE\_DNA\_BEFORE\_PRINTS (medium): Needs both DNA and latent-print work: swab handling zones for touch DNA before powder/cyanoacrylate enhancement; route DNA ↔ Fingerprint in the sequence shown.

Ex-A9 - Mobile phone / SIM (P2, EPI 47.9)

Why: Mobile phone / SIM: relevance 0.85 for homicide / suspicious death (digital evidence), individualizing potential 0.85; context: victim\_link (+0.05); physically stable in storage; irreplaceability 0.80. EPI 47.9.

Handling: Airplane mode if unlocked, otherwise Faraday bag; keep charged; do NOT browse the device; record IMEI.

Does not prove: Device data shows what the device did; attributing actions to a person needs corroboration. Certificate under BSA 2023 §63 needed for admissibility.

Corroborate with: CDR / tower location from service provider; CCTV; Cloud account records

DIGITAL\_VOLATILE (high): Isolate from networks (Faraday bag / airplane mode), keep powered; remote wipe and lock-out risk.

BSA63\_DETAILS\_MISSING (medium): Electronic record: record acquisition method, hash (SHA-256), s.63(2) facts and certificate status — needed to prove it under BSA 2023 s.63.

5. Exhibits held for stage 2

| Exhibit | Type              | EPI  | Reason                                 |
|---------|-------------------|------|----------------------------------------|
| Ex-A16  | Soil / mud sample | 23.8 | Low priority for this crime type (P4). |

6. Gaps and follow-up actions

| Severity | Finding                                                                                                                                                        | Action                                                                                                                                                               |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HIGH     | DNA-bearing exhibits are listed but no reference samples. Scene profiles cannot be compared to victim or suspects without them.                                | Collect reference blood/buccal samples (victim post-mortem; each suspect with consent or court order) and send with the same forwarding letter.                      |
| HIGH     | 10 priority exhibit(s) (4 P1) have no linked in-situ photograph among 5 captions: Ex-A12, Ex-A13, Ex-A11, Ex-A6, Ex-A18, Ex-A15, Ex-A1, Ex-A14, Ex-A10, Ex-A9. | Link each exhibit to its in-situ photograph (scale + exhibit marker) in the photo log before forwarding; courts expect the position to be documented before lifting. |
| MEDIUM   | No fingernail scrapings/clippings from the deceased. If there was a struggle, they may hold the assailant's DNA.                                               | Request the autopsy surgeon to collect scrapings from each hand separately.                                                                                          |
| LOW      | Ex-A3 is a batch of 6 similar exhibits (Cigarette / beedi butt).                                                                                               | Request staged testing: submit the 2-3 exhibits closest to the body / freshest first; test the rest only if the first round is uninformative.                        |

7. Laboratory plan

Recommended policy atc(k=0.5): evidential value retained 96.1% (perishable exhibits 94.6%) vs 96.0% (94.4%) examining everything in listing order and 96.0% for a static score sort; late exhibits 0 vs 0. P1 results in 2.06 working days on average vs 2.38. Stage-1 examinations complete in 11.0 working days; 1 exhibit is held for stage 2 (6 bench-hours not queued ahead of other cases).

| Policy              | Value retained | Perishable value | Late | P1 mean (days) | Makespan (days) |
|---------------------|----------------|------------------|------|----------------|-----------------|
| Pramaan: atc(k=0.5) | 96.1%          | 94.6%            | 0    | 2.06           | 11.0            |
| fifo                | 96.0%          | 94.4%            | 0    | 2.38           | 10.8            |
| priority            | 96.0%          | 94.5%            | 0    | 2.0            | 10.6            |
| status_quo          | 96.0%          | 94.4%            | 0    | 2.38           | 10.8            |

| Division                                    | Examiners | Bench hours | Busy (working days) |
|---------------------------------------------|-----------|-------------|---------------------|
| Fingerprint Bureau (latent prints)          | 2         | 38.5        | 11.0                |
| DNA Profiling                               | 3         | 220.5       | 10.4                |
| Physics (trace, impressions, tool marks)    | 2         | 25.0        | 10.1                |
| Chemistry (fire debris, narcotics, poisons) | 2         | 32.0        | 6.4                 |

| Division                  | Examiners | Bench hours | Busy (working days) |
|---------------------------|-----------|-------------|---------------------|
| Cyber & Digital Forensics | 2         | 32.0        | 3.2                 |
| Toxicology                | 2         | 16.0        | 2.2                 |
| Biology & Serology        | 2         | 9.0         | 1.1                 |

8. Method, limitations and integrity

Priorities were computed by a deterministic, rule-based engine from a versioned forensic knowledge base. For every exhibit the Evidentiary Priority Index (EPI, 0-100) combines probative value (crime-type relevance x individualizing potential x contextual linkage), urgency (estimated hours until evidential quality or the field collection window is lost) and irreplaceability, using weights set per crime type. No language model produces or alters any score, tier or schedule. Degradation half-lives are conservative planning heuristics, not measured constants, and must be calibrated by the receiving laboratory. The ranking is decision support for the Investigating Officer; the FSL may re-order examinations on scientific grounds, and every officer override is recorded in the chain-of-custody ledger.

|                              |                                                                           |
|------------------------------|---------------------------------------------------------------------------|
| Engine version               | 1.0.0                                                                     |
| Knowledge-base SHA-256       | 858d6dd5797dbb56c9a058a840a5c9d933e63a0ef96d31c15271dfe11daab52f          |
| Result SHA-256               | 1c45f08b802ff66c48fdf360be5d05f74b67820a5bfc5ec3fb310c052283673d          |
| Chain-of-custody ledger head | seq 11 - 173d7029ae684592d9396559e4b85d63aa0a53bdf039ac96fdd0a4598f256bc9 |

|                                   |                 |                                                  |
|-----------------------------------|-----------------|--------------------------------------------------|
| Investigating Officer (sign/date) | SHO (sign/date) | Received at FSL - name / date / seals intact Y/N |
|                                   |                 |                                                  |